

Amazon E-Commerce Sales Dashboard

AWS + Power BI | End-to-End Analytics Solution

Arghyajyoti Samui | Data Analytics Portfolio Project | 2021-2024 Dataset

5,000
Total Orders

\$2M
Total Revenue

\$294K
Total Profit

13.1%
Profit Margin %

15.7%
Loss Orders %

Project Overview

An end-to-end business intelligence solution analyzing Amazon e-commerce sales data across 5,000 orders spanning January 2021 to October 2024. Built on AWS cloud infrastructure with Power BI for visualization, this project demonstrates a complete modern data analytics pipeline.

Dataset columns: Order ID, Order Date, Ship Date, Customer, Segment, City, State, Region, Category, Sub-Category, Product, Sales, Quantity, Discount, Profit

Data range: Jan 2021 — Oct 2024 | **Orders:** 5,000 | **Partitioned by:** snapshot_day

Tech Stack

AWS S3 — Data lake storage with auto-partitioning

AWS Glue — ETL pipeline with PySpark scripting

AWS Athena — Serverless SQL query engine

Power BI — Interactive dashboard + DAX measures

ODBC Driver — Athena to Power BI connectivity

Date Table DAX — Time intelligence foundation

Data Architecture Pipeline

1	Raw CSV Upload	Data uploaded to AWS S3 orders/ folder
2	Glue Crawler	Schema auto-discovered and registered in Glue Catalog
3	Glue ETL Job	PySpark job auto-partitions data by snapshot_day
4	AWS Athena	SQL queries run directly on S3 partitioned data
5	ODBC Connection	Power BI connects to Athena via Simba ODBC Driver
6	Power BI Dashboard	Interactive 3-page dashboard with 15+ DAX measures

Dashboard Pages

Page 1	Executive Sales Overview	KPI cards, Revenue & Profit trend, Top 5 Sub-Category, Sales map, Key Analysis
Page 2	Profitability & Operations	Discount vs Profit scatter, Shipping heatmap matrix, Revenue vs Profit clustered bar
Page 3	Growth & Segment Analysis	YoY/MTD/YTD measures, Monthly trend, Segment matrix, Day of Week, Quarterly chart

Key Business Insights

- **15.7%** of all orders generate losses — 0.30 discount is the breaking point
- **Corporate segment** leads profit margin at 13.5% across all years
- **Consumer segment** growing strongest — 12.7% (2021) to 14.0% (2024)
- **Home Office** margin declining — from 13.0% to 12.7% — needs attention
- **Central region** slowest shipping at 3.55 days average
- **Saturday** drives peak revenue at \$154K — 20% above Tuesday
- **2023** was peak year at \$615K driven by lowest average discounts
- **Accessories** tops revenue at \$204K but margin is only 13%

Business Problems Solved

BPS 1	Overall business health KPIs	P1
BPS 2	Category revenue analysis	P1-2
BPS 3	Shipping patterns by region	P2
BPS 4	Discount vs profitability	P2
BPS 6	Segment profit margin	P3
BPS 7	Margin-draining products	P2
BPS 8	YoY and MoM trends	P3
BPS 9	Shipping vs segment correlation	P2

Key DAX Measures Written

Measure	Category	Function Used
Total Revenue	Core	SUMX + (1 - Discount)
Profit Margin %	Core	DIVIDE
Shipping Days	Operations	AVERAGEX + DATEDIFF
Loss Orders Count	Operations	CALCULATE + FILTER + COUNTROWS
Loss Orders %	Operations	DIVIDE
Revenue LY	Time Intelligence	CALCULATE + SAMEPERIODLASTYEAR
Revenue YoY %	Time Intelligence	DIVIDE + LY measure
YTD Revenue	Time Intelligence	TOTALYTD
MTD Revenue	Time Intelligence	TOTALMTD
Revenue MoM	Time Intelligence	CALCULATE + DATEADD